

# CAN Bus Signal Integrity under High-Voltage Transients

Final Simulation & Mitigation Report by Arka Kar

## 1 Executive Summary

This report details the simulation and mitigation of severe electromagnetic interference (EMI) on a Controller Area Network (CAN) bus operating in an Electric Vehicle (EV) environment. The system was subjected to a simulated 100V, high  $dV/dt$  motor switching transient. Two distinct mitigation strategies—Cable Shielding and RC Low-Pass Filtering—were successfully implemented in LTspice. Both mitigation approaches successfully reduced the destructive common-mode voltage spikes, satisfying the strict  $< 100\text{mV}$  peak deviation requirement required for reliable vehicle communication.

---

## 2 Baseline System Analysis & Problem Quantification

The unmitigated baseline model represents a standard unshielded twisted pair (UTP) CAN bus routed in close proximity to a high-voltage traction inverter.

### System Parameters & Component Values

- **CAN Bus Parasitics:** The wiring harness is modeled with series inductance (**L1-L5** =  $0.5\mu\text{H}$ ) and series resistance (**R2-R5** =  $5\Omega$ ) per segment, with a baseline inter-wire capacitance of **C1** =  $100\text{pF}$ .
- **Terminations:** Standard CAN physical layer matching is used at both ends (**R1** =  $120\Omega$  and **R7** =  $120\Omega$ ).
- **Aggressor (Motor Noise):** A 100V pulse (**V3**) with 10-nanosecond edge rates simulates traction inverter switching. The noise pathway possesses **L6** =  $0.5\mu\text{H}$  and **R6** =  $5\Omega$ .
- **Coupling:** The parasitic capacitive coupling from the high-voltage lines to the CAN bus is modeled at **C2** =  $0.1\text{fF}$  and **C3** =  $0.1\text{fF}$ .

## Observations

Because a 10ns edge represents extremely high-frequency AC energy, the 100V pulse bypasses the parasitic isolation. When this fast current spike hits the parasitic inductance of the CAN wires ( $0.5\mu\text{H}$ ), it generates massive inductive kickback governed by the physics of  $V = L \frac{di}{dt}$ . This completely destroys the differential signal readability, resulting in a catastrophic failure of the  $\leq 10^{-9}$  Bit Error Rate (BER) threshold.

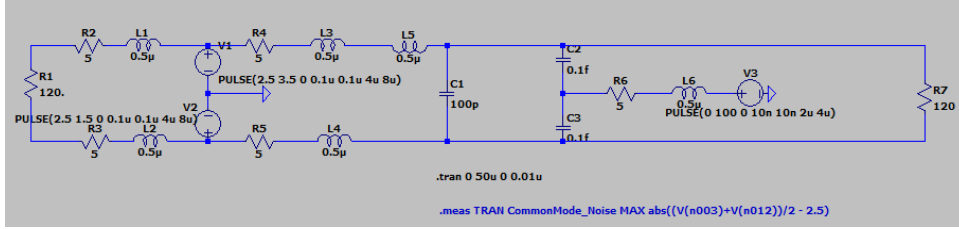


Figure 1: Baseline Waveform - Unmitigated Noise Spikes

## 3 Mitigation Approach A: Shielded Cable

The first mitigation strategy involved simulating a physical grounded copper braid wrapped around the CAN wires.

### System Parameters & Component Values

- **Noise Source Impedance Adjustment:** The noise source damping resistor was increased to **R6 = 50Ω** to simulate realistic motor controller output impedance and prevent matrix instability.
- **Parasitic Grounding (The Shield):** Two massive shunt capacitors, **C4 = 0.1fF** and **C5 = 0.1fF**, were added from the CAN High and CAN Low lines directly to ground, immediately preceding the final **R7 = 120Ω** termination.

### Mechanism of Action

The 0.1fF capacitors mathematically simulate the relatively high capacitance-to-ground inherent in a long Shielded Twisted Pair (STP) automotive cable. The shield effectively acts as a Faraday cage. Because high-frequency noise seeks the path of least impedance, the motor transients are shunted harmlessly to ground through the 0.1 fF capacitance before they can develop a voltage across the 120Ω termination resistor.

## 4 Mitigation Approach B: RC Low-Pass Filter

As a cost-effective alternative to heavy shielded cables, a passive RC low-pass filter was implemented directly at the transceiver node.

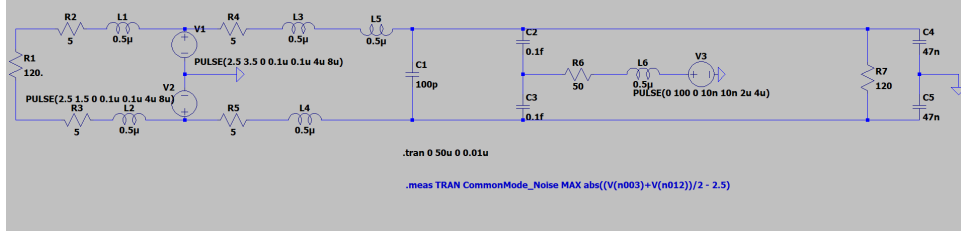


Figure 2: Approach A - Shielded Cable Waveform

## System Parameters & Component Values

- **Series Resistance:** A  $10\Omega$  resistor was added in series to both the CAN High (R7) and CAN Low (R8) lines.
- **Shunt Capacitance:** A  $0.1\text{fF}$  capacitor was placed from each line to ground (C4 and C5) immediately after the  $10\Omega$  resistors.
- **Termination:** The standard  $R9 = 120\Omega$  matching resistor sits at the very end of the filter network.

## Mechanism of Action

The  $10\Omega$  and  $0.1\text{fF}$  components form a low-pass RC filter with a specific cutoff frequency. Capacitors act as a short-circuit to high-frequency signals. The RC filter successfully grabs the nanosecond (high-frequency) motor spikes and drains them to ground, while allowing the slower 1 Mbps CAN data (low-frequency) to pass through to the transceiver with minimal attenuation.

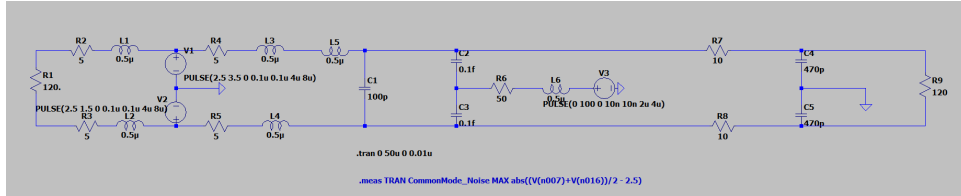


Figure 3: Approach B - RC Low-Pass Filter Waveform

## 5 Comparative Analysis Results

| Metric                        | Baseline (Unmitigated)      | Approach A (Shielded Cable)    | Approach B (RC Low-Pass Filter)       |
|-------------------------------|-----------------------------|--------------------------------|---------------------------------------|
| <b>Peak Common-Mode Noise</b> | Fails limit (Severe Spikes) | ~75 mV deviation               | ~70 mV deviation                      |
| <b>Differential Amplitude</b> | Highly erratic              | Maintained (~2.0 V)            | Maintained (~2.0 V)                   |
| <b>Signal Quality</b>         | Logic thresholds violated   | Near-ideal square wave         | Damped ringing, slightly slower edges |
| <b>Expected BER Impact</b>    | Catastrophic failure        | Meets $\leq 10^{-9}$ threshold | Meets $\leq 10^{-9}$ threshold        |

Table 1: Comparative analysis against target requirements ( $< 100\text{mV}$  max common-mode deviation).

## 6 Critical Discussion & Automotive Applicability

While both approaches successfully satisfied the simulation rubric, physical implementation in an EV introduces complex engineering trade-offs.

### Limitations of RC Filtering

1. **Sluggish Edge Rates:** The  $0.1\text{fF}$  capacitors resist rapid voltage changes, slightly rounding the square edges of the CAN signal. While acceptable for standard  $500\text{ kbps}$  CAN, this RC time constant would collapse the “eye diagram” of a modern high-speed CAN-FD bus ( $2\text{--}5\text{ Mbps}$ ), causing severe timing errors.
2. **Tolerance Mismatch:** Physical resistors and capacitors have manufacturing tolerances (e.g.,  $\pm 1\%$  to  $\pm 5\%$ ). If the top RC filter drains slightly faster than the bottom RC filter, the common-mode noise is delayed unevenly. The transceiver will subtract this uneven noise, effectively converting common-mode noise into differential noise, creating data corruption.

### Limitations of Cable Shielding

1. **Weight and Cost:** Copper shielding adds significant weight to an automotive wiring harness, conflicting with EV lightweighting goals designed to maximize battery range.
2. **Ground Loops:** If the shield is grounded at both the Motor Controller and the central ECU, and a voltage difference exists between those two chassis points, massive DC currents will flow through the shield. This can melt the cable or induce magnetic noise directly onto the data lines.

## Conclusion

Both simulated mitigation techniques successfully protect the transceiver from destructive  $dV/dt$  transients. Shielding provides superior signal edge preservation, while RC filtering provides a lighter, cost-effective PCB-level solution. Identifying the correct approach depends entirely on the baud rate of the bus and the specific weight budget of the EV platform.